MIL-C-36672
1 June 1970

MILITARY SPECIFICATION

CANNULA, TRACHEOTOMY, UNIVERSITY OF KANSAS DESIGN, METAL

This specification is mandatory for use by all Agencies and Departments of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers metal tracheotomy cannulae of the University of Kansas design for use in performing cricothyroidotomy

1.2 Classification. The tracheotomy cannulae shall be of the following sizes as specified (see 6.2).

Size 1 - Adult

Size 2 - Pediatric

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

FSC 6515

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SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
MMM-A-250	Adhesive, Water-Resistant (for Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

Military

MIL-P-116	Preservation, Methods of.

STANDARDS

Federal

FED-STD-151	Metal, Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)
24120 CANNULA, TRACHEOTOMY, UNIVERSITY OF KANSAS DESIGN, METAL

(Copies of specifications, standards, and drawing, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or requests for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F86-68 Recommended Practice for SURFACE REQUIREMENTS OF METALLIC SURGICAL IMPLANTS

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042.)

3. REQUIREMENTS

3.1 Material. All components of tracheotomy cannulae shall be fabricated from type 302 or 304, class 3 material in accordance with Federal Specification GG-I-526. As an alternate, inner tube and outer sleeve shall be fabricated from type 316 or 317, class 3 material in accordance with GG-I-526. All components shall have a final Rockwell hardness of B75 to B85 after proper stress relieving, when tested as specified in 4.4.2.

3.2 Style, design and dimensions. Shall be a tube, Bivalve, University of Kansas design, metal cannula set, suitable for use by technicians with minimum of training in performing cricothyroidotomy. Each tracheotomy set shall consist of a curved cannula member which fits within an outer curved bivalve sleeve member. Inner member shall be concentric and secured to a metal plate at the proximal end designed to permit ease of insertion into and removable from the outer member. The outer bivalve member shall be securely spot-welded to a metal retaining plate at the proximal end. Distal end of inner cannula shall be open and smoothly finished. The curved cannulae (part 1, inner tube shown on drawing) shall be interchangeable with outer sleeve (part 2 on drawing) when tested as specified in 4.4.3. Items shall be designed for use with FSN 6515-935-7091 Dilator-Dagger, Tracheotomy, University of Kansas Design, Curved Blades, 4-1/4 inch Long, Forceps Type, with box lock and FSN 6515-935-7092 Scissors-Dagger, Tracheotomy, University of Kansas Design, Curved Blades, Angulated Pointed Tips, 5-3/16 inch Long. Tracheotomy unit shall be in strict accordance with DPSC Drawing 24120 for appropriate size, stock number and item identification.

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3.3 Concentricity. The maximum allowable total indicated runout (T.I.R.) of inner tube shall be 0.010 inch when tested as specified in 4.4.4.

3.4 Finish. Finish shall be bright (highly reflective) for all components. Definitions of acceptable surface characteristics shall be as described in ASTM F86.

3.4.1 Passivation. All components shall be properly passivated before they are welded or silver-soldered.

3.5 Resistance to boil test, copper sulfate test and saline test. Cannula assemblies shall not show evidence of corrosion when subjected to the boil and copper sulfate tests as specified in 4.4.1.1 and 4.4.1.2, respectively. In addition, tubing of inner tube assembly and split sleeve member of outer sleeve assembly shall not show evidence of corrosion when subjected to the saline test as specified in 4.4.1.3.

3.6 Identification marking. The name or registered trademark of the contractor, and "stainless" shall be permanently and legibly marked on the cannula assemblies.

3.7 Workmanship. Workmanship shall be first class throughout. Tracheotomy cannula assemblies shall be free from defects which detract from their appearance or impair their serviceability, proper functioning, and intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.2 Certificates of quality. Certificates of quality supplied by the manufacturer of the stainless steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table I. Unit of product for examination purposes shall be one cannula.

Table I. Sampling Criteria for Examination

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. Unit of product for tests shall be one cannula.

Table II. Sampling Criteria for Tests

Component	Characteristic	Req.	Test Procedure	Insp level	AQL
Cannula	Hardness	3.1	4.4.1	S-2	1.5
Assembly	Dimensions	3.2	-----	S-2	0.65
	Interchangeability Test	3.2	4.4.3	S-2	0.65
	Total Indicated Runout Test	3.3	4.4.4	S-2	0.65
	Boil Test	3.5	4.4.1.1	S-4	0.65
	Copper Sulfate Test	3.5	4.4.1.2	S-4	0.65
	Saline Test	3.5	4.4.1.3	S-4	0.65

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4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in construction of the tracheotomy cannulae shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The tracheotomy cannulae shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Inspection, however, shall not be restricted to defects listed in Table III. Any cannula in the sample containing one or more defects shall be rejected and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical:	None defined.
Major	
101	Cannula not size indicated.
102	Cannula not complete with all specified component parts.
103	Mating parts fail to fit and/or seat properly into each other.
104	Cannulae not interchangeable.
105	Edges of component parts not free from burrs or sharpness.
106	Components not free from cracks and deformed areas or sections.
107	Solder and spot welded joints not secure and free from spatter or fissures; surrounding surfaces not free from flux, excess solder or discoloration.
108	Distal ends of inner and outer tubes not smooth and properly burnished.
109	Finish of component parts not smooth, uniform and free from roughness.
110	Inner tube and outer sleeve not free from foreign matter.
111	Contractor's name or registered trademark not present or not legible on inner and outer tubes.
112	Labels or printed markings on unit container not correct (particularly stock number and item identificaton).
Minor	
201	Size markings on component parts not present or illegible.
202	Labels or printed markings on unit container illegible and not free from defacing marks, scratches and tears.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Corrosion tests.

4.4.1.1 Boil test. Tracheotomy cannula shall be subjected to the boil test as described and specified in GG-I-526.

4.4.1.2 Copper sulfate test. Tracheotomy cannulae shall be subjected to the copper sulfate test as described and specified in GG-I-526.

4.4.1.3 Saline test. Tracheotomy cannulae shall be scrubbed with soap and warm water, rinsed in hot water, followed by immersion into ethyl alcohol (95 percent) and dried. Cannulae shall then be completely immersed into Normal Saline Solution (U.S.P.) and while immersed, placed into an oven at a temperature of 37 ± 1 degrees Centigrade for a period of 7 days, removed from the oven and examined for corrosion while still immersed.

4.4.2 Hardness test. Rockwell hardness of all components shall be determined in accordance with method 243 of Federal Test Method Standard No. 151.

4.4.3 Interchangeability test. As a means to control continuity of interchangeability, 10 cannulae randomly selected and retained from contractor's last previous shipment to the Government, and properly identified as such, plus 10 samples randomly selected from current shipment (if contractor has not previously supplied the item to the Government, 20 samples from the current shipment shall be randomly selected) shall be tested as follows:

> Inner cannulae and outer sleeves shall be completely separated. The inner cannulae and outer sleeves shall be reassembled, making certain that no inner cannula is mated with the same outer sleeve from which it was removed. The inner cannula shall be capable of moving in and out of the outer sleeve without evidence of objectionable sticking, and shall seat completely in the outer sleeve as indicated on the drawing.

4.4.4 Concentricity test. The outside surfaces of the inner tubes shall be tested for T.I.R. in the following manner:

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Remove the inner tube from the outer sleeve. A suitable measuring instrument to measure outside diameters (O.D.) of the inner tubes shall be used to measure the O.D. of the tubes at three places: at the most distal end, near the most proximal end, and at the approximate center portion of the tube length. The measurements shall be recorded and the difference between the minimum measurement and the maximum measurement shall be recorded.

4.4.5 Retention samples. Three cannula assemblies from those selected to perform the interchangeability test specified in 4.4.3 shall be forwarded to the Defense Personnel Center, Attention: DPSC-ATQ, for retention. Retention samples shall be marked with the following information:

Retention samples from interchangeability test,
Federal Stock Number, Size,
Contract Number, Date Submitted,
Name of Contractor

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One cannula, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	144 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

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5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material.

5.1.3.4 Unit package. Each unit shall be packaged in a polyethylene bag heat-sealed at both ends.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MIL-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tracheotomy cannulae described herein shall be used in performing cricothyroidotomy. All inner tubes and outer sleeves shall be interchangeable with each other within a given size.

6.2 Ordering data. Procurement document should specify:

(a) Number, title and date of this specification.
(b) Size (see 1.2).
(c) Applicable Federal stock number.
(d) Applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers the following items listed in the Federal Supply Catalog:

Federal Stock Number	Item Identification
6515-935-7089	CANNULA, TRACHEOTOMY, UNIVERSITY OF KANSAS DESIGN, Metal, Pediatric.
6515-935-7090	CANNULA, TRACHEOTOMY, UNIVERSITY OF KANSAS DESIGN, Metal, Adult.

Cutodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1476

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-36672 FSN:

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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